NeuropathVision: A Knowledge Graph-Enhanced LLM Agent for Patient-Centered Interpretation of Neuropathology Reports

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***Abstract*—Neuropathology reports are written for clinicians, making them difficult for patients to understand. Recent advances in large language models (LLMs) suggest a path to plain-language explanations for lay readers. In this study, we present NeuropathVision, a knowledge graph-enhanced LLM agent that generates patient-centered interpretations of neuropathology reports. The agent combines semantic retrieval from authoritative sources with a glioma-focused domain knowledge graph capturing entities and relations. It applies prompt-level style constraints such as clarity, transparent uncertainty, and empathy. To evaluate NeuropathVision, a nine-item patient-centered readability assessment form was developed covering four dimensions: Accuracy, Clarity, Empathy & Respect, and Treatment & Prognosis. Twenty cases were rated independently by two senior neuropathologists. All interpretations met the prespecified pass criteria, and the pooled mean score was 96.78. Exact agreement on item-level ratings was 86.7%. Clarity and empathy scores were high, reflecting LLM strengths in following style guidance. The knowledge graph contributed structured anchors and consistency checks that supported accuracy. These results indicate the feasibility and acceptability of NeuropathVision for producing patient-centered neuropathology interpretations. Future work will add statement-level citations and evaluate patient-reported outcomes.**

***Index Terms*—patient-centered pathology report, large language model, knowledge graph, neuropathology, health communication.**

I. Introduction

Pathologic diagnosis is the gold standard for cancer identification, and pathology reports provide the diagnostic reasoning that informs treatment and prognosis [1]. These reports integrate histopathologic, molecular, and clinical evidence into statements that guide clinical decision-making. However, they are originally designed for professional audiences such as pathologists, oncologists, and surgeons rather than for patients [2]. The language is highly specialized, compressed, and often relies on implicit reasoning. Ambiguous phrasing, variable diagnostic certainty, and limited explanation can cause inconsistent understanding even among specialists [3]. Patients therefore often find reports overly technical and struggle to interpret their diagnoses [4].

Previous research has primarily aimed to improve professional communication to mitigate diagnostic misinterpretation. Miscommunication may occur even among clinicians because of inconsistent terminology, differing staging or grading systems, and inadequate specification of uncertainty [5]. To address these challenges, structured reporting standards and computational tools have been developed to harmonize terminology and improve consistency in diagnostic documentation [6]. For instance, PathChat integrates multimodal data to assist in professional summarization of pathology reports [7]. Although these models have demonstrated improved readability and semantic alignment for clinical users, they remain optimized for expert audiences. They do not directly address the needs of patients who lack medical training and require explanations that are both comprehensible and emotionally appropriate.

The widespread implementation of electronic health records and patient portals has transformed the way patients access their diagnostic information. Many patients now view their pathology results online or through mobile applications shortly after release. Surveys indicate that most patients prefer immediate access to their results because it promotes transparency, engagement, and a sense of control over their health [8]. However, when reports contain uncertain descriptive language, patients often experience confusion and anxiety while waiting for clarification from their physicians [9]. Studies on patient perspectives highlight that the combination of greater accessibility but lower interpretability can lead to misperception, unnecessary distress, and reduced trust in clinical communication [10]. This pattern underscores an urgent need for systems that can generate timely, accurate, and patient-centered explanations of pathology results.

Recent advances in large language models (LLMs) offer new opportunities to transform complex biomedical text into plain and patient-friendly language. LLMs have demonstrated remarkable fluency in summarizing and simplifying medical information [11]. LLMs such as GPT-4 [6] and Med-PaLM 2 [12] have been applied in various clinical language understanding tasks, including pathology or radiological report extraction and lay summary generation [13]. However, the direct use of LLMs in medical communication presents chal-

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lenges. LLMs may produce factual inconsistencies, semantic drift, or hallucinated statements, particularly when handling domain-specific terminology. Also, OCR errors in scanned pathology documents can increase the risk of misinterpretation and spread misinformation. Prior work shows that clinically reliable medical text generation should be grounded in structured and trusted knowledge sources, such as ontologies or knowledge graphs (KGs), to support clinical soundness and transparency [14] [15].

To address these limitations, we propose NeuropathVision, a KG-enhanced LLM agent that delivers patient-centered interpretations of neuropathology reports, with gliomas as a representative use case. NeuropathVision build a structured neuropathology KG from the latest World Health Organization (WHO) Classification of Tumors of the Central Nervous System (CNS) [16] and the National Molecular Diagnostic Guidelines [17]. It uses graph-based retrieval-augmented generation (RAG) to align outputs with verified clinical source, and patient-centered prompts to produce clear explanations of histopathologic findings, molecular results, and integrated diagnostic summaries. Our main contributions are as follows:

- We construct a glioma-focused and structured KG that encodes WHO taxonomy, key molecular markers, standard neuropathology terminology, and expert diagnostic heuristics. This KG provides an authoritative structure for accurate and explainable report interpretation.
- We develop a KG-based RAG pipeline with patient-centered prompting. Graph rules handle entity linking, guidline- and style-based constraint checks, and conflict detection, to keep interpretations clinically consistent and accessible to patients.
- We implement an end-to-end interpretation agent. Two senior neuropathologists independently reviewed its outputs to assess clinical trustworthiness, clarity for lay readers, and transparency.

II. Methods

This study presents NeuropathVision, a KG-enhanced, end-to-end LLM agent that generates patient-centered interpretations of neuropathology reports. As shown in Figure 1, the workflow includes report preprocessing, dual-path knowledge extraction and retrieval, patient-centered prompt construction, and LLM generation of clinically accurate, lay-friendly, and empathic interpretation.

A. Input Preprocessing

Neuropathvision accepts reports as photos or typed text. For photo inputs, OCR is applied to recover the full text, followed by de-identification, normalization, and segmentation into a semi-structured record. Key fields include patient sex and age, specimen or site, histopathologic diagnosis, grade or stage, and relevant immunohistochemistry markers when available. The semi-structured record provides a consistent interface for knowledge extraction and retrieval. For typed inputs, the same field extraction and schema validation are applied. Dates and identifiers are removed prior to downstream processing.

B. Knowledge Extraction

NeuropathVision derives two complementary representations from the semi-structured report text. One path performs text slicing and vectorization to obtain dense semantic vectors for fast similarity search. The other performs triplet extraction to produce normalized subject-relation-object facts aligned with the domain KG. These paths jointly support unstructured retrieval and graph-based reasoning.

1) Chunking and Embedding: To enable rapid semantic retrieval, the pathology text is first chunked into fine-grained semantic chunks. This leverages the report's semi-structured format to preserve completeness of context. The resulting chunks are then embedded by a domain-adapted encoder. Chunks may be single sentences or short marker-focused fragments (e.g., "*IDH1 R132H (+)*"). Chunking granularity is tuned to balance semantic coherence with retrieval accuracy, avoiding both overly coarse segments and over-fragmentation.

For embedding, we use a medical-domain pretrained model. PubMedBERT [18] serves as the encoder that converts each text chunk $s_i \in \mathcal{S} = \{s_i, \cdots, s_n\}$ into a fixed-dimensional semantic embedding $v_i^s \in \mathbb{R}^d$.

These embeddings support rapid semantic retrieval against a local structured database, enabling efficient matching between the input and similar chunks. The database in this study is built by applying the same chunking and encoding procedure (using PubMedBERT) to relevant corpora v_i^c, including guidelines, textbooks, and peer-reviewed literature (see Section II-C1).

2) Knowledge triplet extraction: Medical named entity recognition (MNER) and relation extraction (RE) are performed based on pathology texts. MNER identifies key entities from $\mathcal{D} = \{d_1 \cdots d_n\}$, while RE infers semantic relations among predefined types $\mathcal{T} = \{t_1 \cdots t_k\}$. We adopt a weakly supervised pipeline. First, an NER model fine-tuned on Transformers-sklearn toolkit [19] detects core entities $\mathcal{E} = \{(e_j, e_k) | e_j, e_k \in \mathcal{D}\}$ such as disease types (e.g., glioma) and gene or protein markers (e.g., *IDH1*).

Building on these entities, a distant-supervision and multi-instance learning (MIL) framework extracts relations between entity pairs. Let $B_{j,k} = \{d_i | (e_j, e_k) \in d_i\}$ denote the bag of texts containing the pair (e_j, e_k). The MIL classifier g_{MIL} predicts the relation $r_{j,k}$ using bag-level features,

$$r_{j,k} = g_{\text{MIL}}\left(\bigcup_{d_i \in B_{j,k}} \phi(e_j, e_k, d_i);\ \mathcal{R}_{\mathcal{D}} \right) \tag{1}$$

where $\mathcal{R}_D$ is the set of candidate relation labels derived from the knowledge base (see Section II-C1).

This process yields normalized triples such as *"IDH1-has mutation-R132H"* and *"ATRX-has mutation-nuclear expression loss"*, which serve as structured inputs for subsequent retrieval and augmentation.

C. Dual-Path Knowledge Retrieval

The retrieval layer combines semantic vector retrieval for unstructured evidence (Section II-B1) with KG augmentation

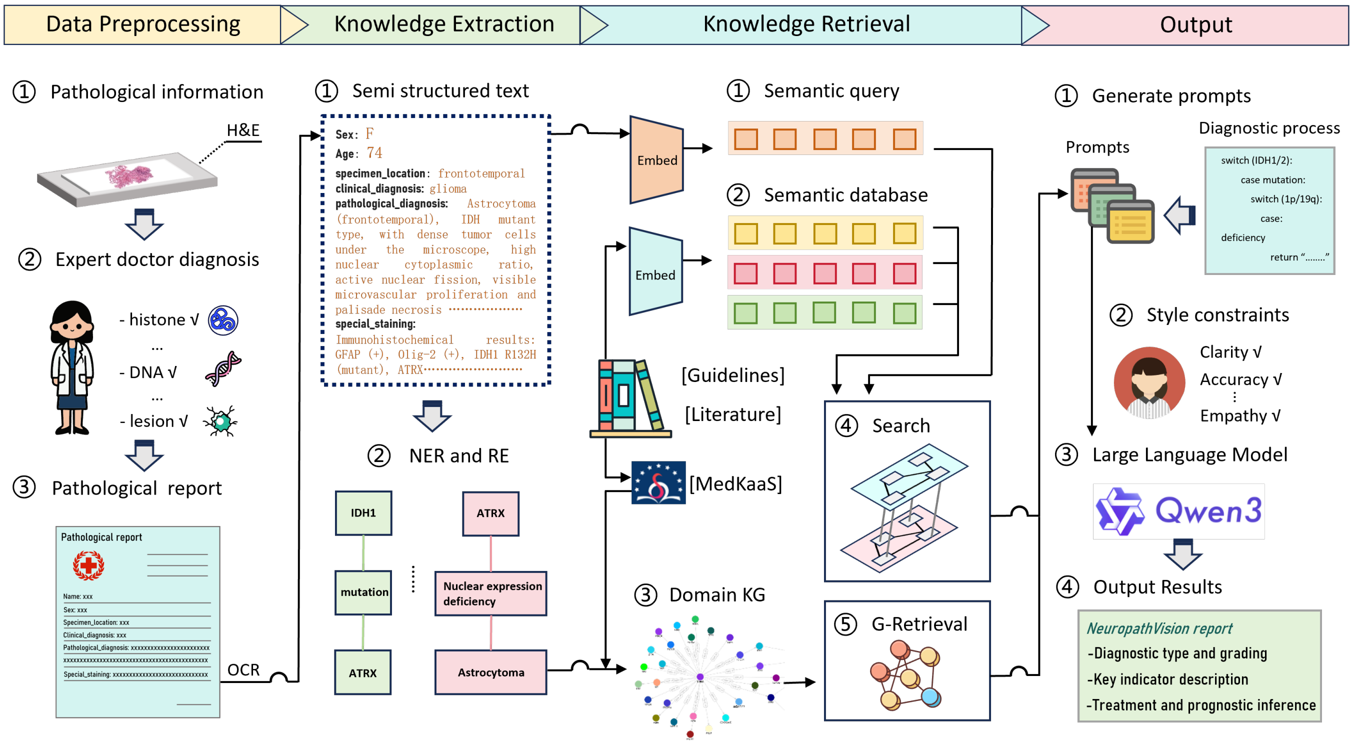

Fig. 1: Overview of the NeuropathVision workflow. (a) Scanned reports are converted into semi-structured text via OCR and standardized formatting; (b) PubMedBERT embeddings and Transformer-based MNER with MIL extract key entities and relations; (c) Evidence is retrieved by vector similarity search and graph-based reasoning over a glioma-specific KG built on MedKaaS [30] using graph neural networks and G-Retrieval; (d) Retrieved evidence is merged into prompts and fed to Qwen3 to generate the final interpretation.

for structured reasoning (Section II-B2). This dual-path strategy enriches each case with authoritative domain knowledge and preserves traceability for prompt construction.

1) External knowledge curation and vector-based retrieval: We curated authoritative glioma pathology sources, including the WHO CNS tumor classification, specialist textbooks, and peer-reviewed PubMed literature. Domain experts cleaned the sources and segmented them into semantically complete fragments, grouped by topics such as diagnostic criteria, genetic marker interpretation, and prognostic assessment. This process preserves content integrity while removing redundancy before embedding and retrieval. We used PubMedBERT [18] to encode both the curated fragments and the sliced input report into vector representations, which were stored in a local vector database (see Section II-B1). At inference, a graph neural network [20] returns the top-N fragments most similar to the case vectors, providing efficient and precise access to relevant unstructured evidence.

2) Domain KG retrieval: Vector retrieval is complemented by a domain-specific KG constructed from glioma guidelines. The graph covers core entities such as glioma subtypes (e.g., astrocytoma, glioblastoma) and genetic or protein markers (e.g., *IDH1, ATRX*), connected by relations including mutation associations and pathologic features. Given the case entities $\mathcal{E}$ and extracted triplets (e_j, r, e_k), a graph-retrieval procedure selects relevant subgraphs as follows:

$$\mathcal{G}_{\text{output}} = \bigcup_{k \in Q} \left\{ \underset{\mathcal{G}_k \subseteq \mathcal{G}_0}{\text{argmax}} \left(\sum_{(e_j, r, e_k) \in T_k} f_\theta(e_j, r, e_k) \right) \right\}. \quad (2)$$

Here $\mathcal{G}_0$ denotes the domain KG, $\mathcal{G}_k$ is a candidate subgraph centered on query entity k, T_k is the set of triplets associated with k, and f_θ is a scoring function. The method identifies clinically meaningful chains such as "*IDH1-mutation-low grade glioma*," and extracts compact subgraphs that capture marker-grade relationships. These structured information is then combined with unstructured text snippets retrieved by vector search.

D. Patient-Centered Prompt Design for NeuropathVision

The task prompt for NeuropathVision integrates a curated evidence set with instruction templates and patient-centered style constraints, including clarity, plain language with first-mention definitions, explicit treatment of uncertainty, an empathic tone, and balanced option framing. The prompt is iteratively refined with expert feedback to guide the Qwen3 model. At inference time, the self-constructed domain KG for gliomas constrains reasoning toward clinically accurate outputs. The final deliverable is a structured interpretation that summarizes like diagnosis, key molecular markers, and prognosis. Periodic clinical reviews support continuous updates to

[Prompts main structure]

You are a senior neuropathologist. I need to handle the following glioma related pathological reports:
1. Please provide an interpretation based on the patient's pathology report.
2. Strictly interpret according to the "Chinese Guidelines for Integrated Diagnosis and Treatment of Tumors - Gliomas - Diagnosis and Evaluation" to ensure the accuracy and rigor of the information.
3. Use easy to understand language to enable patients to comprehend the content of the report; Maintain a positive attitude and highlight the information that patients are most concerned about.
4. Adjust the report content according to the patient's age to better fit their individual situation.
5. The template for the template is as follows (note that only the template content is output without additional explanation)

[Template content]

1. Tumor type and grade
{Specific content, requirements: professional, accurate, easy to understand}
2. Key test results
{Specific content, only interpret positive indicators, only interpret undisputed indicators, requirements: professional, accurate, easy to understand}
3. Suggestions for you:
{The specific content of "suggested content" should be written from the aspects of adjusting mentality, following medical advice, cooperating with treatment, and recovery, without using strongly recommended words such as "as soon as possible", "as much as possible", "strongly", etc. The requirement is to be able to help patients and full of humanistic care。}

The medical team will develop the most suitable plan for you based on the final test results. Please maintain close communication with the attending physician!

[NeuroPathVision output]

Semantic slicing of patient pathology report content (Chunk):
{Age: 67}
{Specimen location: right frontal lobe and right insula}
{Clinical diagnosis: intracranial space occupying lesions, symptomatic epilepsy [secondary epilepsy]}
{Immunohistochemical results: IDH1 R132H (wild-type), IDH2 R172K (wild-type), EGFR (+).}
{Pathological diagnosis:Right frontal and right insula: glioblastoma, NOS (unspecified), active tumor nuclear division, visible microvascular proliferation and palisade necrosis, invading the submucosa
{Histopathological grade: CNS WHO 4}

Knowledge retrieval
{If IDH wild-type astrocytoma is accompanied by EGFR amplification, chromosome 7 gain/10 deletion, or TERT promoter mutation (typical molecular features of glioblastoma), even in the absence of necrosis and/or microvascular proliferation, it is diagnosed as "glioblastoma, IDH wild-type, WHO grade 4".}

Integrative pathological diagnosis of glioma KG:

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Pseudo Code:
functional gliomas diagnoise:
  switch (IDH1/2):
    switch (IDH1/2):
    Case wild:
      If H3 wild type:
        If microvascular hyperplasia | | necrosis | | TERT promoter mutation | |+7/-10
chromosome mutation | | EGFR
amplification:
  Glioblastoma, IDH wild-type......
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Fig. 2: English versions of NeuropathVision's patient-centered prompt templates and output rules, including retrieval chunking and graph-based constraints.

retrieval resources and prompt design. An English translation of the prompt template, including extracted inputs, retrieval chunking, graph-based rules, is shown in Figure 2.

III. Evaluation Protocol

A. Design and Participants

We conducted a preregistered evaluation of NeuropathVision's patient-centered interpretations on de-identified neuropathology cases diagnosed as gliomas. For each case, we prepared the signed histopathology report and the corresponding patient-centered interpretation generated by NeuropathVision. Two senior neuropathologists independently rated all cases. Case order was randomized for both raters. Raters had access to the signed report only for checking concordance and were blinded to each other's scores.

B. Patient-Centered Readability Assessment Form

We used a nine-item patient-centered readability assessment form with four dimensions: Accuracy, Clarity, Empathy & Respect, and Treatment & Prognosis. The dimensions and items were adapted from established standards in cancer reporting and health communication [21]. Accuracy means item-level concordance with the signed report with no additions, exaggerations, or omissions across diagnosis, grade, and molecular findings. This is consistent with protocols from the College of American Pathologists [22]. Clarity places the main message first, uses a well-structured layout, employs plain language with definitions at first mention, and favors concise phrasing in active voice. This aligns with validated health-literacy tools such as the Agency for Healthcare Research and Quality Patient Education Materials Assessment Tool [23]. Empathy & Respect uses a supportive and non-judgmental tone, makes uncertainty explicit, and outlines follow-up actions. This follows the oncology communication guidances [24] [25] [26]. Treatment & Prognosis presents options neutrally and in a balanced way, communicates probabilities transparently, and expresses prognosis as ranges rather than absolute claims. This reflects risk-communication standards [27] [28].

Each item was scored on a five-point Likert scale, where one indicates strongly disagree and five indicates strongly agree. The nine items are summarized in Table I.

C. Evaluation Metrics

For each case, the total score was computed as the mean of the nine item scores multiplied by twenty, yielding a range from zero to one hundred. An interpretation was considered acceptable if the total score was at least eighty and each dimension's average score was at least four [29]. Inter-rater agreement between the two neuropathologists was assessed using the exact agreement at the item and dimension levels.

IV. Evaluation Results

Following the protocol in Section 3, two senior neuropathologists independently rated twenty patient-centered interpretations.All evaluations were performed within the NeuropathVision application, and the user interface is shown in Figure 3. All twenty met the prespecified pass criteria, so the observed pass rate was 100%. The lowest total score was 81. Mean total scores were 96.44±2.81 for Rater 1 and 97.11±3.15 for Rater 2 (mean±standard deviation). The pooled mean total score across raters was 96.78±2.93. Across the 180 paired item-level Likert ratings, exact agreement was 86.7%.

Dimension-wise results were Accuracy 4.15, Clarity 4.95, Empathy & Respect 5.00, and Treatment & Prognosis 4.85. Ratings for Clarity and Empathy & Respect were near five, which is consistent with the designed prompt style constraints. Accuracy and Treatment & Prognosis were relatively lower while still exceeding the priori threshold. Item level variation concentrated in Item 1 and Item 8, with exact agreement of about 50% on each, reflecting occasional omissions relative to the signed report and a preference for more conservative prognosis phrasing with explicit uncertainty.

V. Discussion

NeuropathVision generated patient-centered interpretations that met the prespecified pass criteria in all twenty glioma

TABLE I: Patient-centered readability assessment form for NeuropathVision.

Item No.	Dimension	Item
1	Accuracy	Matches the signed pathology report on diagnosis, grade or stage, and molecular findings with no additions, no exaggerations, and no omissions.
2	Clarity	Places the main message first and uses a well-structured layout.
3		Explains technical terms at first mention or provides lay synonyms.
4		Uses short sentences and mainly active voice and avoids long sentences and acronym overload.
5	Empathy & Respect	Uses supportive and respectful language and avoids alarming, cold, or blaming wording.
6		States uncertainty explicitly and provides a follow-up plan.
7	Treatment & Prognosis	Presents therapy and drug options neutrally and avoids definitive or suggestive recommendations.
8		Expresses prognosis or survival as ranges with uncertainty and avoids absolute claims.
9		Summarizes main pros and cons and applicability in plain language to support discussion with clinicians.

cases. Ratings for Clarity and for Empathy & Respect were near ceiling. Consistent with prior work, these results suggest that carefully designed prompts help LLMs follow style constraints and generate fluent plain-language explanations [31].

By contrast, ratings for Accuracy and for Treatment and Prognosis were relatively lower. Most deductions in Accuracy arose from omissions of details that were present in the signed report. Despite these omissions, the Accuracy dimension still averaged above four, which indicates a stabilizing effect from the constructed domain KG. The KG mirrors glioma reporting with explicit entities and relations. It supplied structured anchors for diagnosis, grade or stage, and key markers. It also provided consistency checks that reduced outright errors when the generative model skipped minor details [32].

Pathologists prefer conservative language for prognosis that states uncertainty clearly and defers individualized estimates to the clinician. It is consistent with current discussions about responsibility for text produced with LLM agents [33]. In settings where clinical teams remain accountable for patient-facing content, cautious phrasing helps avoid overconfidence or misleading statements [27].

This pilot study has limitations. The sample is modest and drawn from a single center. We assessed expert ratings of interpretations rather than patient comprehension, affect, or downstream outcomes. Thus, patient-level impact remains to be established.

In conclusion, NeuropathVision met a predefined acceptability standard in all cases. The KG contributed to accuracy through structured evidence and internal consistency checks,

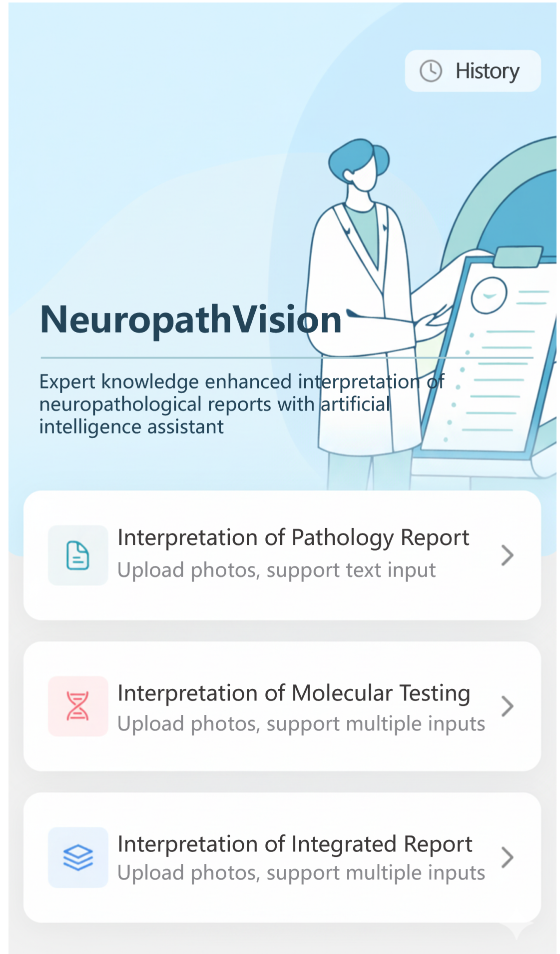

Fig. 3: NeuropathVision interface in English.

while the language model supported clarity and an empathic tone. Future work will expand the dataset and rater pool, extend the graph beyond glioma, run ablations on retrieval and prompting, add claim-level provenance linking each statement to a source, and include patient-reported outcomes and prospective measures of communication quality.

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REFERENCES

[1] M. Mossanen, L. D. True, J. L. Wright, F. Vakar-Lopez, and J. L. Gore, ”Surgical pathology and the patient: a systematic review evaluating the primary audience of pathology reports,” *Human Pathology*, vol. 191, no. 11, pp. e152-e153, 2014.

[2] M. Mossanen, J. L. Wright, L. True, F. Vakar-Lopez, D. Lavallee, and J. L. Gore, ”MP15-20 surgical pathology and the patient: A systematic review evaluating the primary audience of pathology reports,” *J. Urol.*, vol. 191, no. 4S, pp. e152–e153, Apr. 2014.

[3] E. Steimetz, E. Mostafidi, C. Castagna, R. Gupta, and R. Frasso, ”Forgotten clientele: A systematic review of patient-centered pathology reports,” *PLoS ONE*, vol. 19, no. 5, p. e0301116, May 2024.

[4] G. Lazaro, "When positive is negative: Health literacy barriers to patient access to clinical laboratory test results," *J. Appl. Lab. Med.*, vol. 8, no. 6, pp. 1133–1147, Nov. 2023.

[5] H. Kaur, H. Gabriel, M. O. Awiwi, E. Maheshwari, C. Lopes Vendrami, T. Konishi, M. W. Taggart, M. Magnetta, L. C. Kelahan, and S. Lee, "Anatomic basis of rectal cancer staging: Clarifying controversies and misconceptions," *Radiographics*, vol. 44, no. 7, p. e230203, Jul. 2024.

[6] D. Truhn, C. M. L. Loeffler, G. Müller-Franzes, S. Nebelung, K. J. Hewitt, S. Brandner, K. K. Bressem, S. Foersch, and J. N. Kather, "Extracting structured information from unstructured histopathology reports using generative pre-trained transformer 4 (GPT-4)", *J. Pathol.*, vol. 262, no. 3, pp. 310–319, Mar. 2024.

[7] M. Y. Lu, B. Chen, D. F. K. Williamson, R. J. Chen, M. Zhao, A. K. Chow, K. Ikemura, A. Kim, D. Pouli, A. Patel, et al., "A multimodal generative AI copilot for human pathology", *Nature*, vol. 634, no. 8033, pp. 466–473, Jul. 2024.

[8] B. D. Steitz, R. W. Turer, C.-T. Lin, S. MacDonald, L. Salmi, A. Wright, C. U. Lehmann, K. Langford, S. A. McDonald, T. J. Reese, et al., "Perspectives of patients about immediate access to test results through an online patient portal", *JAMA Netw. Open*, vol. 6, no. 3, p. e233572, Mar. 2023.

[9] M. H. Keshavarzi, S. Safaie, S. A. A. Faghihi, and S. Zare, "Barriers of physician-patient relationships in professionalism: A qualitative study", *J. Adv. Med. Educ. Prof.*, vol. 10, no. 3, pp. 199-207, Jul. 2022.

[10] H. Hong and H. J. Oh, "The effects of patient-centered communication: Exploring the mediating role of trust in healthcare providers", *Health Commun.*, vol. 35, no. 4, pp. 502–511, Apr. 2020.

[11] E. Yu, X. Chu, W. Zhang, X. Meng, Y. Yang, X. Ji, and C. Wu, "Large language models in medicine: Applications, challenges, and future directions", *Int. J. Med. Sci.*, vol. 22, no. 11, pp. 2792-2805, Nov. 2025.

[12] K. Singhal, T. Tu, J. Gottweis, R. Sayres, E. Wulczyn, M. Amin, L. Hou, K. Clark, S. R. Pfohl, H. Cole-Lewis, et al., "Toward expert-level medical question answering with large language models", *Nat. Med.*, vol. 31, no. 3, pp. 943–950, Mar. 2025.

[13] V. Patwardhan, D. Balchander, D. Fussell, J. Joseph, A. Joshi, H. Troutt, J. Ling, K. Wei, B. Weinberg, and D. Chow, "Leveraging large language models to enhance radiology report readability: A systematic review", *J. Am. Coll. Radiol.*, 2025.

[14] E. Steimetz, J. Minkowitz, E. C. Gabutan, J. Ngichabe, H. Attia, M. Hershkop, F. Ozay, M. G. Hanna, and R. Gupta, "Use of artificial intelligence chatbots in interpretation of pathology reports", *JAMA Netw. Open*, vol. 7, no. 5, p. e2412767, May 2024.

[15] X. Yang, Y. Xiao, D. Liu, Y. Zhang, H. Deng, J. Huang, H. Shi, D. Liu, M. Liang, X. Jin, et al., "Enhancing doctor-patient communication using large language models for pathology report interpretation", *BMC Med. Inform. Decis. Mak.*, vol. 25, no. 1, p. 36, Jan. 2025.

[16] D. N. Louis, A. Perry, P. Wesseling, D. J. Brat, I. A. Cree, D. Figarella-Branger, C. Hawkins, H. K. Ng, S. M. Pfister, G. Reifenberger, et al., "The 2021 WHO classification of tumors of the central nervous system: A summary", *Neuro Oncol.*, vol. 23, no. 8, pp. 1231–1251, Aug. 2021.

[17] F. Sahm, S. Brandner, L. Bertero, D. Capper, P. J. French, D. Figarella-Branger, F. Giangaspero, C. Haberler, M. E. Hegi, B. W. Kristensen, et al., "Molecular diagnostic tools for the World Health Organization (WHO) 2021 classification of gliomas, glioneuronal and neuronal tumors; an EANO guideline", *Neuro Oncol.*, vol. 25, no. 10, pp. 1731–1749, Oct. 2023.

[18] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, "Domain-specific language model pretraining for biomedical natural language processing", *ACM Trans. Comput. Healthc.*, vol. 3, no. 1, pp. 1–23, Jan. 2021.

[19] F. Yang, X. Wang, H. Ma, and J. Li, "Transformers-sklearn: A toolkit for medical language understanding with transformer-based models", *BMC Med. Inform. Decis. Mak.*, vol. 21, no. S2, pp. 90, 2021. https://doi.org/10.1186/s12911-021-01459-0

[20] H. Ootomo, A. Naruse, C. Nolet, R. Wang, T. Feher, and Y. Wang, "Cagra: Highly parallel graph construction and approximate nearest neighbor search for GPUs", in *Proc. IEEE 40th Int. Conf. Data Eng. (ICDE)*, pp. 4236–4247, Apr. 2024.

[21] Centers for Disease Control and Prevention (U.S.). Office of the Associate Director for Communication. (2013). CDC clear communication index : a tool for developing and assessing CDC public communication products : user guide. https://www.cdc.gov/ccindex/pdf/clear-communication-user-guide.pdf. Accessed 19 Dec. 2025.

[22] College of American Pathologists. *Cancer protocols*. CAP. https://www.cap.org/protocols-and-guidelines/cancer-reporting-tools/cancer-protocols. Accessed 19 Dec. 2025.

[23] S. J. Shoemaker, M. S. Wolf, and C. Brach, "Development of the Patient Education Materials Assessment Tool (PEMAT): A new measure of understandability and actionability for print and audiovisual patient information", *Patient Educ. Couns.*, vol. 96, no. 3, pp. 395–403, Sep. 2014.

[24] T. Gilligan, N. Coyle, R. M. Frankel, D. L. Berry, K. Bohlke, R. M. Epstein, E. Finlay, V. A. Jackson, C. S. Lathan, C. L. Loprinzi, et al., "Patient-clinician communication: American Society of Clinical Oncology consensus guideline", *J. Clin. Oncol.*, vol. 35, no. 31, pp. 3618–3632, Nov. 2017.

[25] W. F. Baile, R. Buckman, R. Lenzi, G. Glober, E. A. Beale, and A. P. Kudelka, "SPIKES—a six-step protocol for delivering bad news: Application to the patient with cancer", *Oncologist*, vol. 5, no. 4, pp. 302–311, Jul. 2000.

[26] F. Stiefel, C. Bourquin, P. Salmon, L. A. Achtari, S. Dauchy, N. Ernstmann, L. Grassi, Y. Libert, F. Vitinius, D. Santini, et al., "Communication and support of patients and caregivers in chronic cancer care: ESMO Clinical Practice Guideline", *ESMO Open*, vol. 9, no. 7, p. 103496, Jul. 2024.

[27] P. Abhyankar, R. J. Volk, J. Blumenthal-Barby, P. Bravo, A. Buchholz, E. Ozanne, D. C. Vidal, N. Col, and P. Stalmeier, "Balancing the presentation of information and options in patient decision aids: An updated review", *BMC Med. Inform. Decis. Mak.*, vol. 13, no. S2, p. S6, 2013.

[28] B. Fischhoff, *Communicating Risks and Benefits: An Evidence Based User's Guide*. Washington, DC, USA: U.S. Government Printing Office, 2012.

[29] S. H. Javid, M. R. Kilgore, E. J. Austin, E. U. Parker, R. Alvarez, M. R. Flanagan, E. G. Brewer, C. Gibbons, S. K. Holt, J. M. Lee, et al., "The development and comparative effectiveness of a patient-centered pathology report for breast cancer care: A randomized clinical trial", *Support. Care Cancer*, vol. 33, no. 3, pp. 248-260, Mar. 2025.

[30] X. Xu, X. Wang, M. Wu, H. Ma, L. Shen, and J. Li, "Development of an interactive medical knowledge graph based tool set", *Proc. Comput. Sci.*, vol. 221, pp. 578–584, 2023.

[31] J. W. Ayers, A. Poliak, M. Dredze, E. C. Leas, Z. Zhu, J. B. Kelley, D. J. Faix, A. M. Goodman, C. A. Longhurst, M. Hogarth, et al., "Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum", *JAMA Intern. Med.*, vol. 183, no. 6, pp. 589–596, Jun. 2023.

[32] L. Tang, Z. Sun, B. Idnay, J. G. Nestor, A. Soroush, P. A. Elias, Z. Xu, Y. Ding, G. Durrett, and J. F. Rousseau, "Evaluating large language models on medical evidence summarization", *NPJ Digit. Med.*, vol. 6, no. 1, p. 158, Sep. 2023.

[33] D. O. Shumway and H. J. Hartman, "Medical malpractice liability in large language model artificial intelligence: Legal review and policy recommendations", *J. Osteopath. Med.*, vol. 124, no. 7, pp. 287–290, Jul. 2024.